

Warning System For OVER-HEIGHT VEHICLES

NAVPREET SINGH TUNG

This circuit can be used to detect big over-height vehicle moving towards bridges, tunnel entrances and overhead structures, and warn the drivers. It can be developed for monitoring such a vehicle approaching, say, a bridge at night.

The system uses an infrared (IR) sensor to monitor the vehicle. An audio-visual alarm is activated when

the over-height vehicle is detected by the system.

For example, a truck loaded with hazardous gases colliding with an overhead bridge can be dangerous. If a vehicle is too high to safely pass under the bridge, a sensor triggers an alarm placed at a distance ahead on the route. The alarm warns the driver to alter the course or stop before col-

liding with the overhead structure.

Circuit and working

The circuit diagram of the vehicle detection system is shown in Fig. 1. It has an Arduino Uno board, IR proximity sensor module, 16x2 LCD (LCD1), piezo buzzer, preset (10-kilo-ohm), 9V battery and LED (LED1).

IR proximity module. This module incorporates an IR transmitter and an IR receiver/detector in one package (Fig. 2). It transmits an IR beam continuously. Whenever an object or over-height vehicle is detected, the IR beam reflected from the object/vehicle is detected by the inbuilt IR receiver.

When the reflected IR beam is detected, the system generates an audio-visual alert informing the driver to stop the vehicle, as it may not be safe to continue. LED1 is used as a visual alert whenever an over-height vehicle is detected. Sensing range is adjustable with an inbuilt variable resistor in the IR module.

Arduino Uno. Arduino Uno that has an ATmega328 microcontroller (MCU) is used in the project. It consists of 14 digital input/output (I/O) pins, six analogue inputs, 16MHz crys-

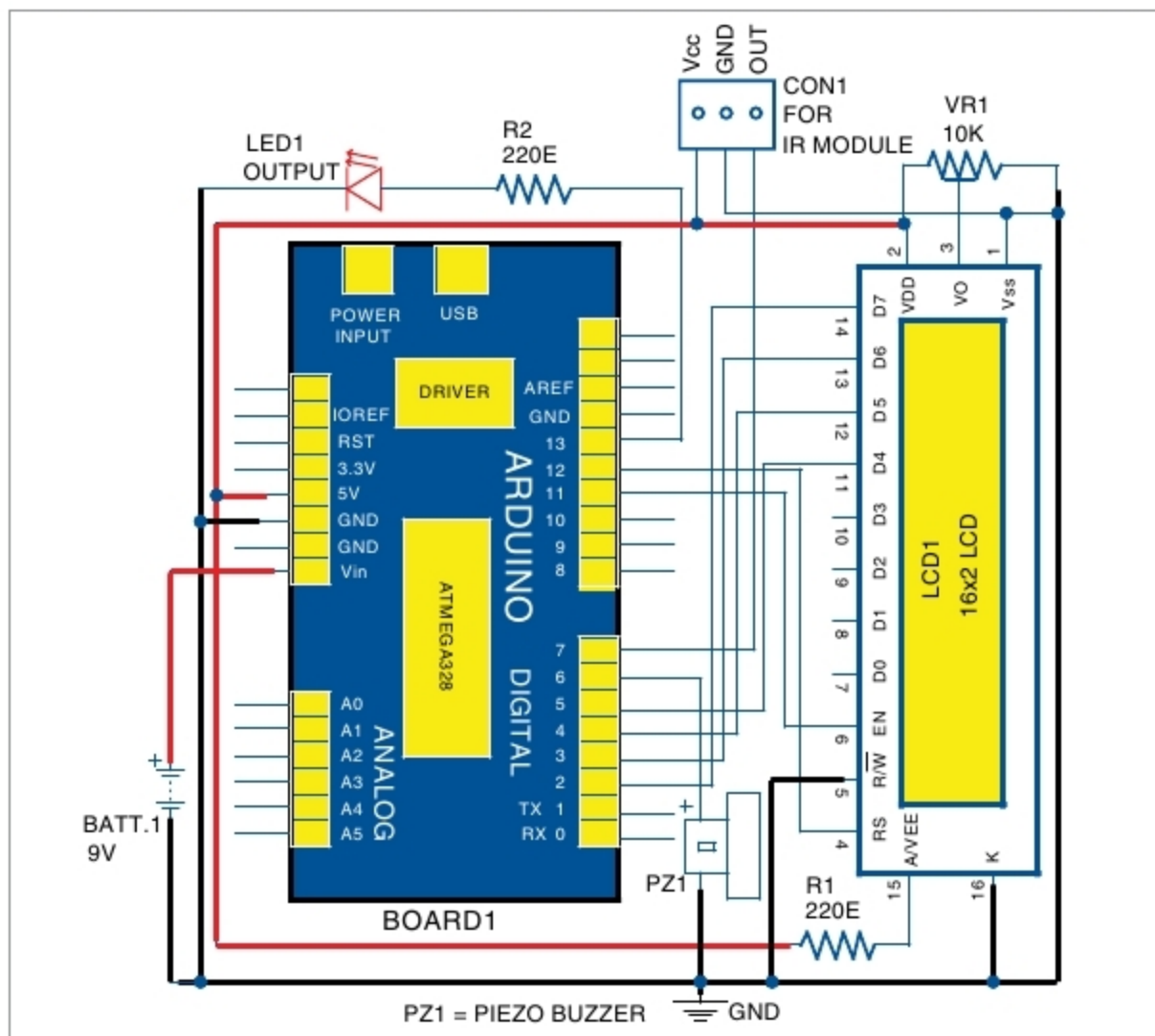


Fig. 1: Circuit diagram of vehicle detection system

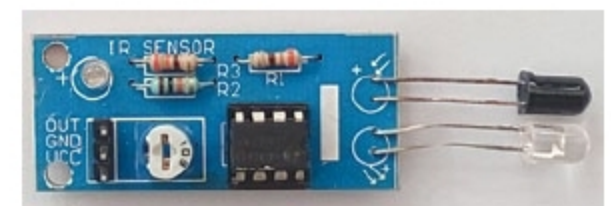


Fig. 2: IR proximity module

THE COMPLETE MAGAZINE ON OPEN SOURCE

OpenSource
ForYou

The latest from the Open Source world is here.

OpenSourceForU.com

Join the community at facebook.com/opensourceforu

Follow us on Twitter @OpenSourceForU